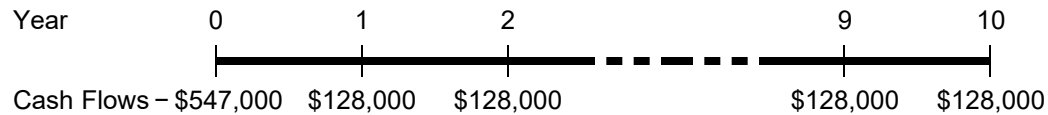


Worked Solution (Formula Solution)

a. What is the payback period of this project?

Here is the timeline for the project:



The payback period is:

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Cash Flow}}$$

Therefore,

$$\text{Payback Period} = \frac{\$547,000}{\$128,000} = 4.27 \text{ years}$$

The payback period is 4.27 years.

b. Should you take the project if you want to increase the value of the company?

The net present value of the project is:

$$NPV = \frac{CF}{r} \times \left[1 - \frac{1}{(1+r)^n} \right] - \text{Initial Investment}$$

Therefore,

$$NPV = \frac{\$128,000}{0.114} \times \left[1 - \frac{1}{(1+0.114)^{10}} \right] - \$547,000 = \$194,343.32$$

If you want to increase the value of the company, you will take the project since the NPV is positive.

